Hüseyin Felek



Software Engineer | Linux Platforms, Networking & Automation

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SUMMARY

Engineer with 4.5+ years designing, deploying and operating Linux-based platforms in production, from an embedded Linux automation controller platform with custom device drivers and OTA infrastructure running on 1000+ field devices to an OpenWrt-based networking platform. Broad hands-on experience across automation and scripting (Bash, Python), containerization, release and fleet management, device log collection and web-based monitoring. Served as the technical escalation point for field issues, leading root cause analysis and working with internal teams and external vendors to restore and improve reliability.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Stabilized core L2/L3 features (DHCP relay, IEEE 802.1X/RADIUS, ARP inspection, IGMP) on an OpenWrt-based Linux switching and routing platform, debugging each feature to production readiness.
- Root-caused reliability and throughput issues through packet-level analysis with Wireshark; extended SNMP/MIB support to improve network monitoring.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Incident Handling and Root Cause Analysis

- Served as the technical escalation point for customer and field issues, running root cause analysis and coordinating fixes with customers and the field service team.
- Coordinated integration and issue resolution with an international PROFINET stack vendor, and led post-production quality assurance for the controller platform.

Linux Platform and Deployment

- Architected an embedded Linux automation controller platform end to end: Linux customization, device drivers, service configuration and device lifecycle management, including runtime integration of the CODESYS automation runtime.
- Owned image and release management for the device fleet: built and versioned device images, maintained changelogs, and ran backup, verification and restore processes, delivered through a file-based OTA update system.
- Containerized on-device services and embedded databases with Docker, with custom images and desktop and mobile tools for deploying images and backups to field devices.
- Redesigned the controllers' I/O data exchange from full-cycle to change-based updates and rolled it out to 1000+ field devices, replacing legacy controllers.
- Implemented power-backup logic to protect devices and data against unexpected shutdowns.

Monitoring, Automation and Tooling

- Built web-based interfaces for device configuration, monitoring and OTA updates, and implemented log collection from field devices to support diagnostics and troubleshooting.
- Wrote Bash and Python scripts for device control, diagnostics and test automation; developed REST APIs and an I/O simulator for hardware-free testing, and delivered device data over MQTT.
- Managed Git-based build and release workflows with Makefiles (GitLab/GitHub-hosted repositories) for software and firmware; developed STM32/FreeRTOS firmware with a secure OTA bootloader (SBSFU).

CORE COMPETENCIES

Linux & Platforms	Embedded Linux customization, device drivers, Linux service and process management (systemd), containerization (Docker), OpenWrt, Linux CLI
Scripting & Programming	Bash, Python, C/C++, Makefile
Build, Release & Tooling	Git, GitLab, GitHub, image and release management, OTA deployment, backup and restore, REST APIs (FastAPI, Postman), Jira, Confluence
Networking & Troubleshooting	TCP/IP, L2/L3, VLAN, IEEE 802.1X/RADIUS, ACL, Wireshark, root cause analysis
Monitoring & Data	Grafana, SNMP monitoring, log collection, MQTT, SQL

EDUCATION & LANGUAGES

B.Sc., Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate